

Data Science for Software Engineering

Assignment # 3

Objectives:

Analyze the impact of TF-IDF versus one-hot encoding on matrix decomposition (PCA/SVD) and their effectiveness in supervised classification tasks using K-Nearest Neighbors (KNN) and Logistic Regression, with a focus on dimensionality reduction and model evaluation via cross-validation.

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First, think about statement problems and then write your program.
2. Write the Program in Python/IDE and save a notebook.
3. Complete your assignment **within the given deadline**.
4. Please submit your **files** using this naming convention: ROLLNO_SECTION_AssNO.
5. Submit your Assignment on Google Classroom.

TASKS

The dataset is attached to the assignment.

Task 1: TF-IDF Feature Extraction

1. Parse Risk Factors, Symptoms, and Signs, and subtype from stringified lists to Python lists..
2. Convert each list into a single string (e.g., ["fever", "stress"] → "fever stress").
3. Apply TF-IDF vectorization to each column (Risk Factors, Symptoms, Signs) separately.
4. Use TfidfVectorizer from sklearn.feature_extraction.text.
5. Combine the TF-IDF matrices into a single feature matrix.
6. Compare the TF-IDF matrix with the provided one-hot encoded matrix (e.g., sparsity, unique features).

Task 2: Dimensionality Reduction

1. Apply PCA and Truncated SVD to both matrices (TF-IDF and one-hot).
 - Reduce dimensions to 2–3 components.
 - Compare the explained variance ratio between TF-IDF and one-hot encoding.
2. Visualize the reduced dimensions (2D plots).
 - Color-code diseases by category (e.g., cardiovascular, neurological).
 - Discuss which encoding method produces more separable clusters.

Task 3: Train KNN Models on both TF-IDF and one-hot encoded matrices:

1. Use $k = 3, 5, 7$ and three distance metrics:
 - Euclidean
 - Manhattan
 - Cosine similarity
2. Perform 5-fold cross-validation for each k and metric combination.
3. Report the following metrics for each configuration:
 - Accuracy
 - Precision
 - Recall
 - F1-score
4. Train Logistic Regression models on both matrices:
 - Perform 5-fold cross-validation.
 - Compare performance (accuracy, F1-score) against KNN.
5. Compare results across:
 - Encoding methods (TF-IDF vs. one-hot).
 - Distance metrics (for KNN).
 - Model types (KNN vs. Logistic Regression).

Task 4: Critical Analysis

Write a report addressing:

- Why TF-IDF might outperform one-hot encoding (or vice versa) for this dataset.
- Clinical relevance of the results (e.g., "Do TF-IDF clusters align with real-world disease categories?").
- Limitations of both encoding methods.

Deliverables:

Ipynb file with complete implementation and code should be divided into cells according to tasks and subtasks.

Streamlit Application for KNN.

Blog, and LinkedIn Post.

(Best of luck 😊)